AGI

Benchmarks
Feasibility
Implications

Frederik Schittny | Institute of Theoretical Informatics, KIT
16.02.2026


Sam Altman
@sama
In from three to eight years we will have a machine with the general intelligence of an average human being
8:11 PM · Mar 31, 2025 · 5.5M Views
2K
4K
58K
3.4K

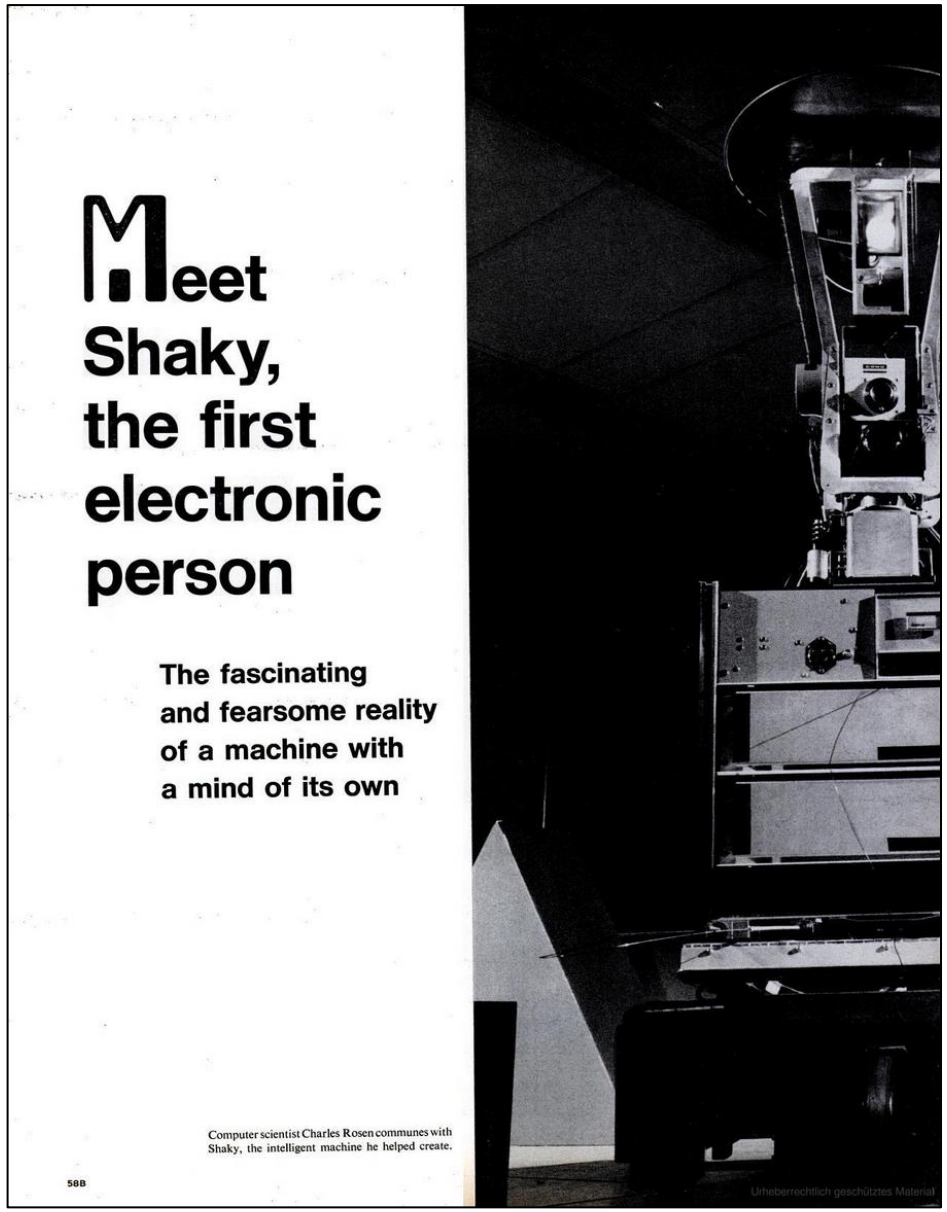

Marvin Minsky
@marv_minsky
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– Marvin Minsky, *Life* Magazine, November 1970

[0][Life Magazine, 1970]



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Structure

- AGI Definition – *Ambiguous Foundations*
- Benchmarks – *Some Proposals*
- Feasibility – *Timelines and Approaches*
- Implications – *Propagating Inaccuracy*
- Discussion

AGI

[1][Russel & Norvig, 2020] [2][Hendrycks et al., 2025] [3][Hutson, 2025]

AGI

Artificial

[1][Russel & Norvig, 2020] [2][Hendrycks et al., 2025] [3][Hutson, 2025]

AGI

Artificial General

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Defining Intelligence (is really hard)

[2][Hendrycks et al., 2025] [3][Hutson, 2025]

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internal processes

behavior

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[2][Hendrycks et al., 2025] [3][Hutson, 2025]

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The most important question to keep in mind when talking about AGI:

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What is the underlying definition of AGI in this context?

The Turing Test

- Indistinguishability in conversation[4]
  - Beat to various degrees
  - Language models “specialized” for this test
- Criticism
  - Can be “faked” easily[5]
  - Language bad indicator for intelligence[6, 7]



[xkcd 329]

[4][Turing, 1949] [5][Weizenbaum, 1966] [6][Fedorenko et al., 2025] [7][The Verge, 2025]

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Underlying intelligence understanding:

Indistinguishable (language) between human and AI



[xkcd 329]

[4][Turing, 1949] [5][Weizenbaum, 1966] [6][Fedorenko et al., 2025] [7][The Verge, 2025]

Humanity's Last Exam

- Intended for foundation models
- Traditional LLM benchmarks "too easy"
  - Solution: make the questions harder
- 2500 expert level questions
  - Mostly scientific
  - "Multimodal" (text and images)

Classics

Question:



Here is a representation of a Roman inscription, originally found on a tombstone. Provide a translation for the Palmyrene script. A transliteration of the text is provided: RGYNᵓ BT ḤRY BR ᶜTᵓ ḤBL

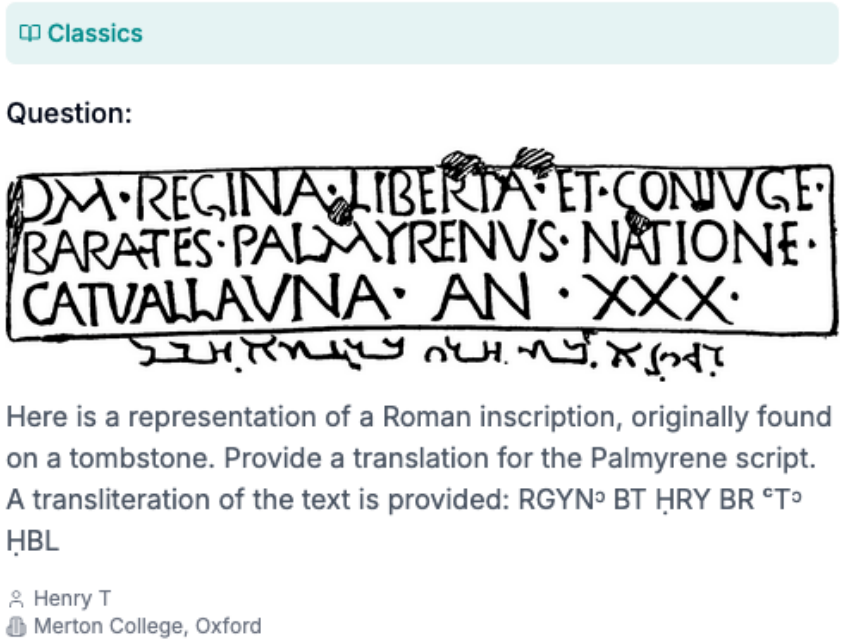
Henry T
Merton College, Oxford

[lastexam.ai]

[8] [Phan et al., 1949]

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[lastexam.ai]

Underlying intelligence understanding:

Expert level knowledge and reasoning skills across many topics

[8] [Phan et al., 1949]

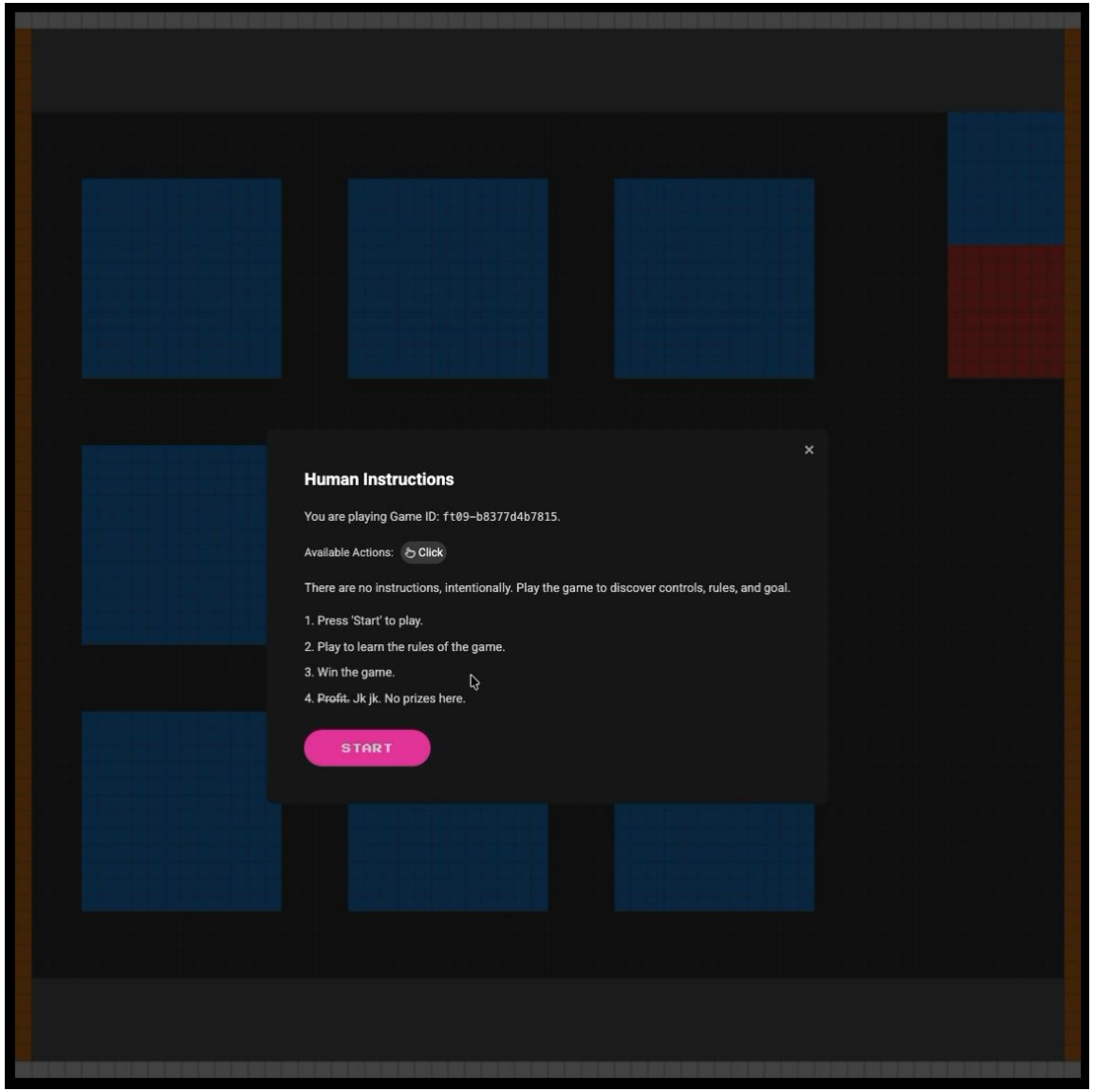
ARC-AGI

- Measuring fluid intelligence
- Static puzzles
  - Somewhat comparable to IQ tests
  - Symbolic, compositional and contextual reasoning
- Interactive puzzles
  - Small games (without given rules)
  - Goal acquisition, planning, exploration

[9][Chollet et al., 2025] [10][arcprize.org]

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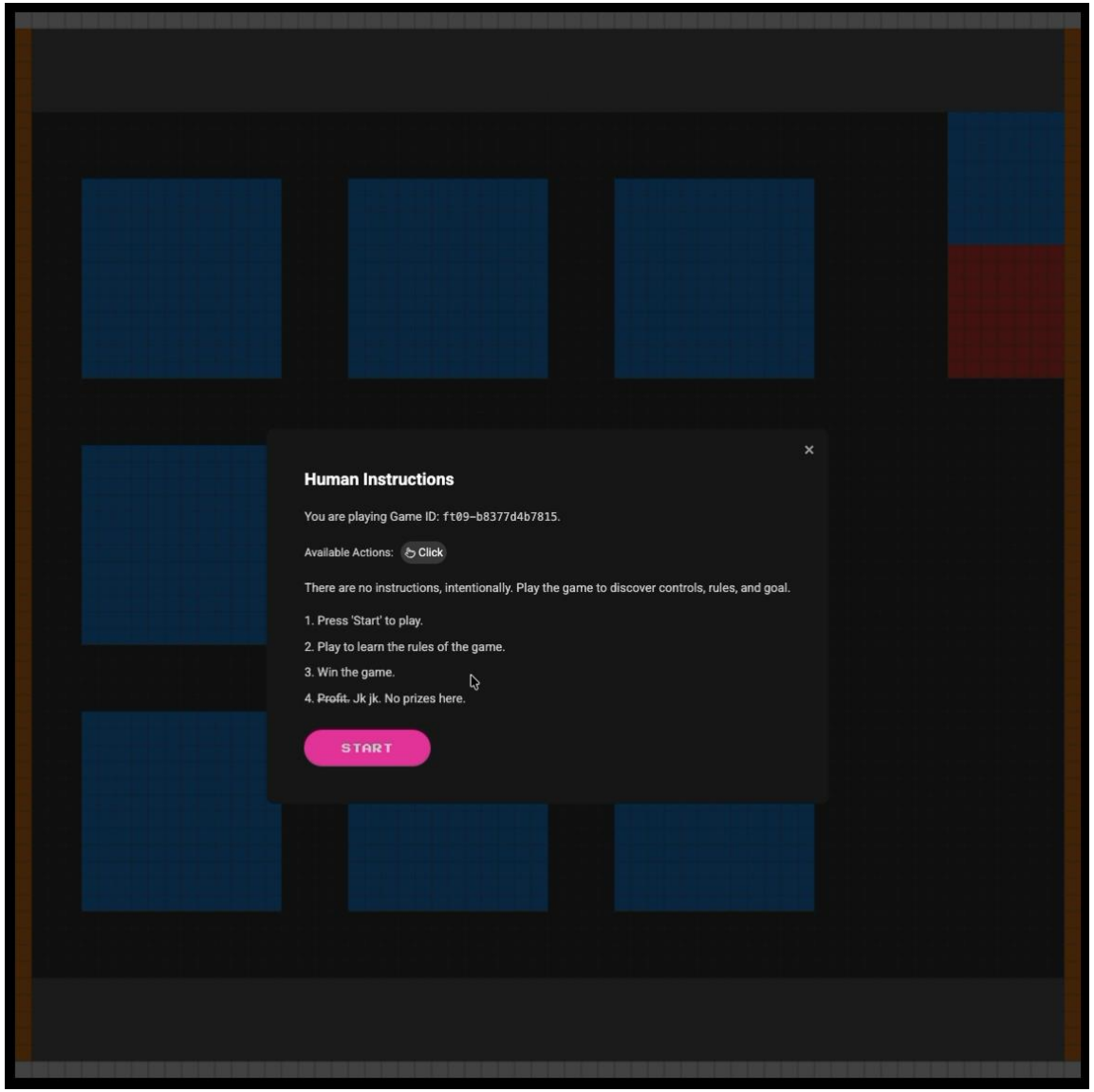
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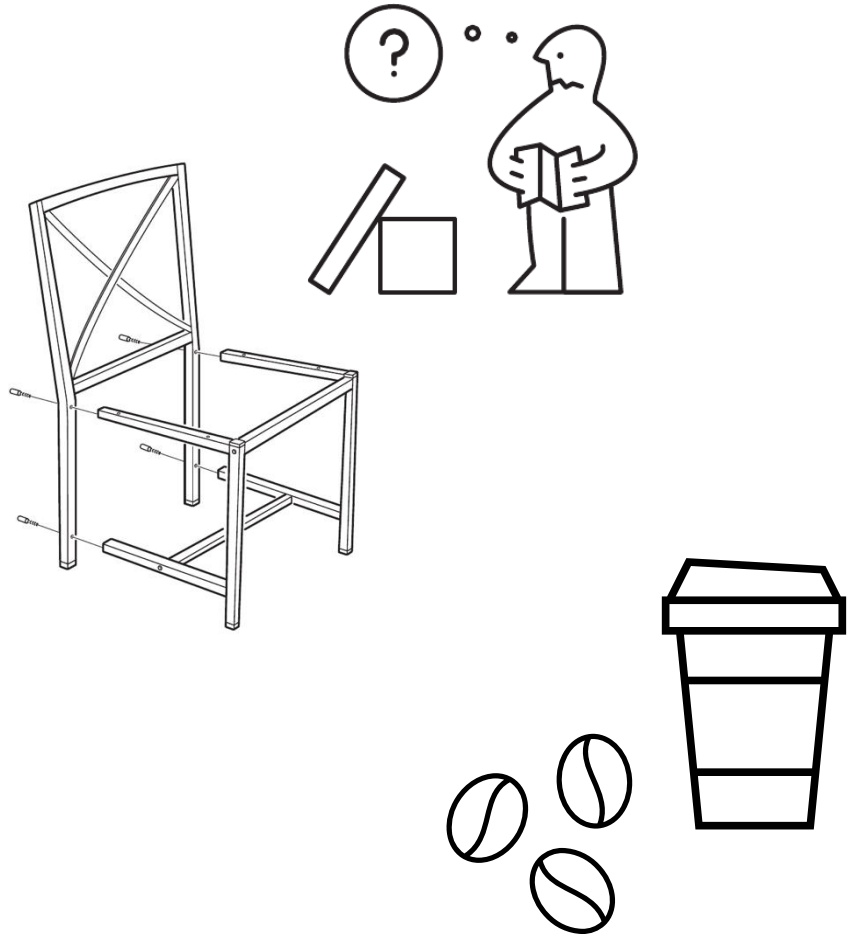
Ability to adapt, learn, and generalize



[9][Chollet et al., 2025] [10][arcprize.org]

Physical Turing Tests

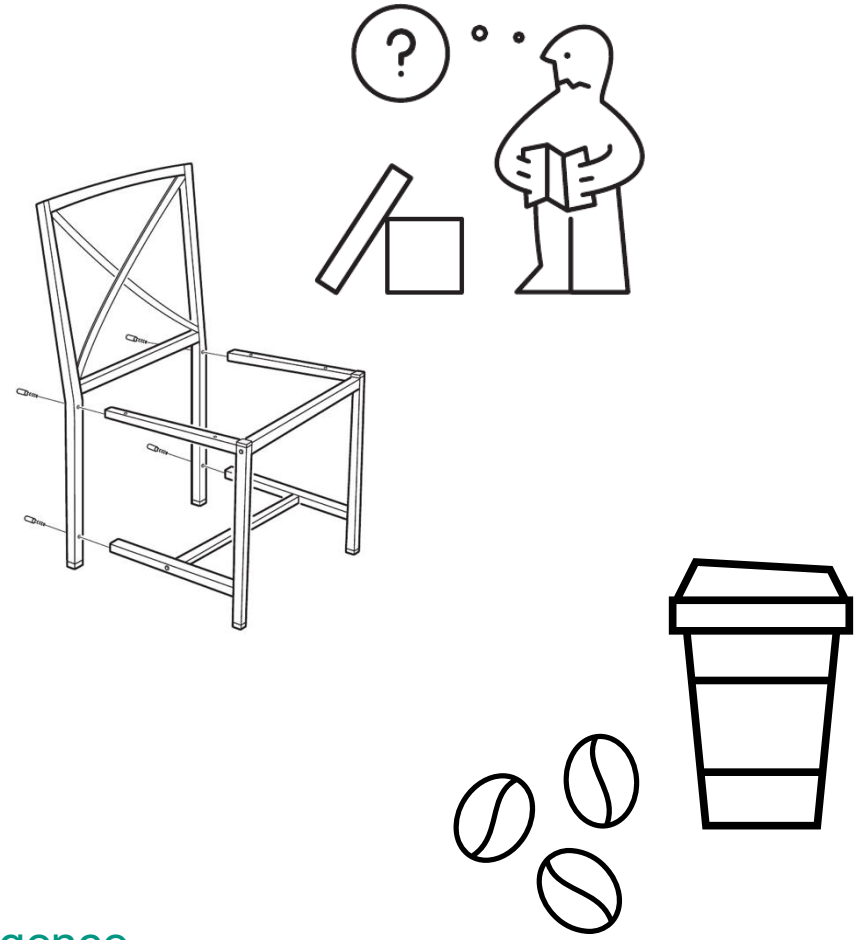
- Incorporating robotics and embodied AI
- Physical goal fulfillment
  - IKEA test [11]
  - Coffee test [12]
- Physical indistinguishability
  - The Physical Turing Test [13]



[11][Vice, 2015] [12][Goertzel et al., 2012] [13][Fan, 2025] [14][Boston Dynamics, 2021]

Physical Turing Tests

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  - Coffee test [12]
- Physical indistinguishability
  - The Physical Turing Test [13]



Underlying intelligence understanding:

Physical presence and interaction as crucial part of intelligence

[11] [Vice, 2015] [12] [Goertzel et al., 2012] [13] [Fan, 2025] [14] [Boston Dynamics, 2021]

Benchmarking AGI

- Many different approaches
- Defined by underlying understanding of intelligence
- Specialized for certain AI subfields/modalities
  - Can be combined
  - But no guarantee for completeness
- No harmonized approach without shared definition

Feasibility - Timelines

When will AGI be reached?

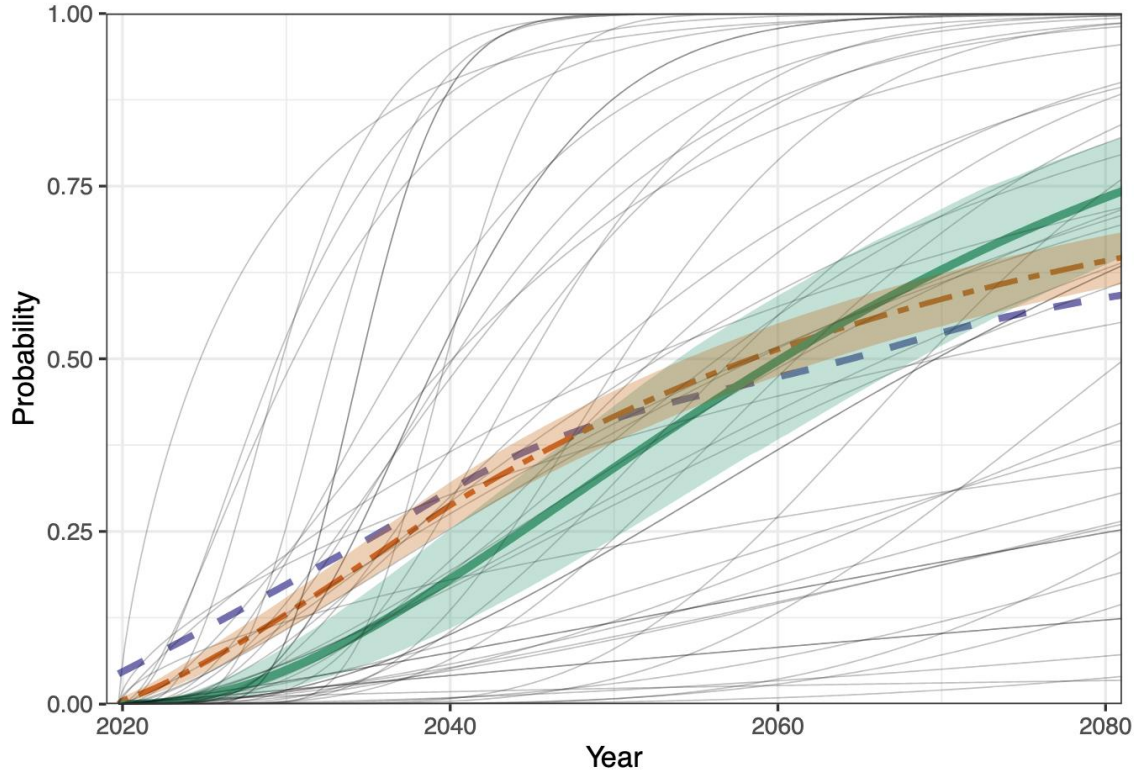
Feasibility - Timelines

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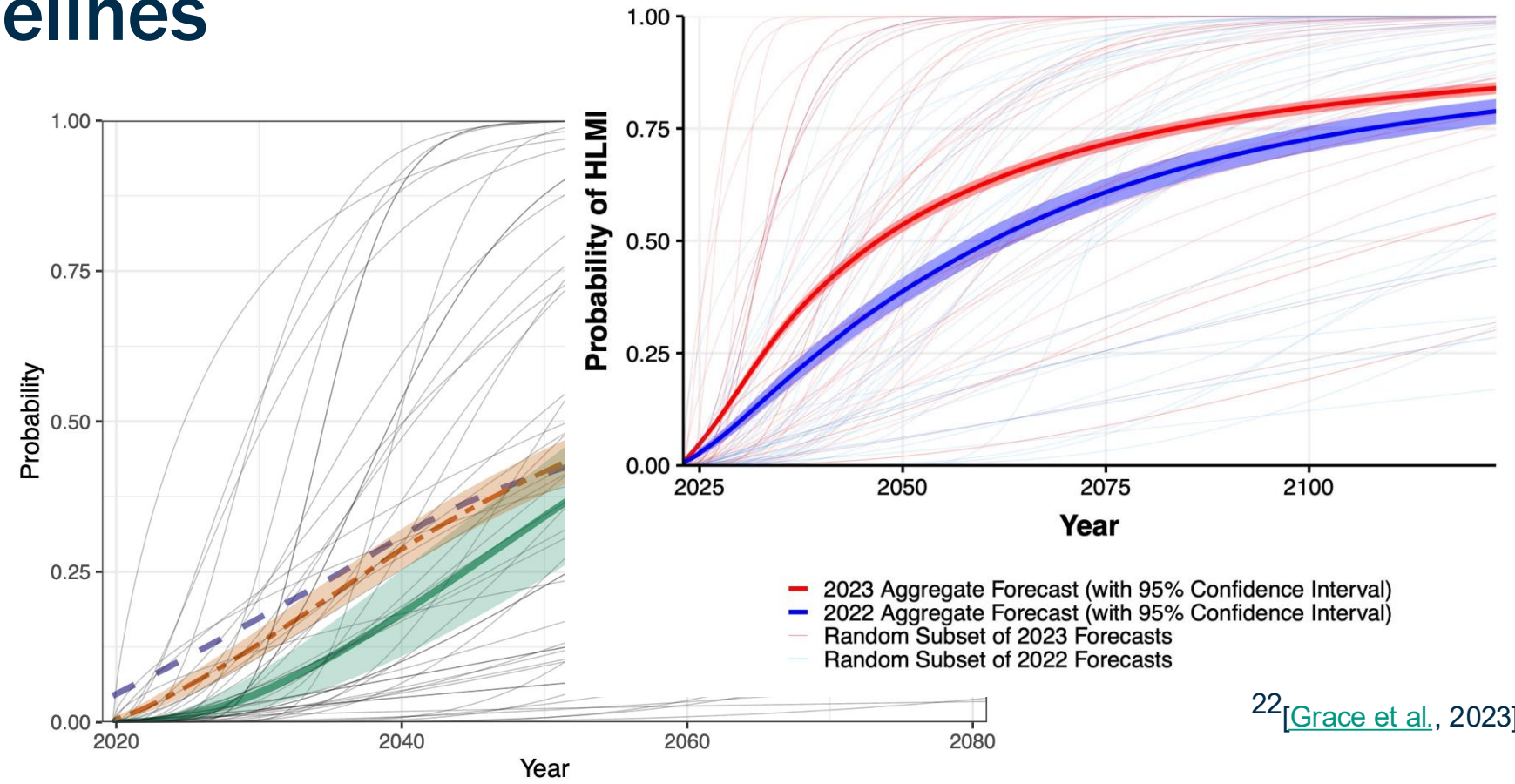
[15][Futurism, 2024] [16][Agüera y Arcas & Norvig, 2023] [17][Futurism, 2023] [18][Business Insider, 2025] [19][Dreyfus, 1992] [20][Fjelland, 2020]

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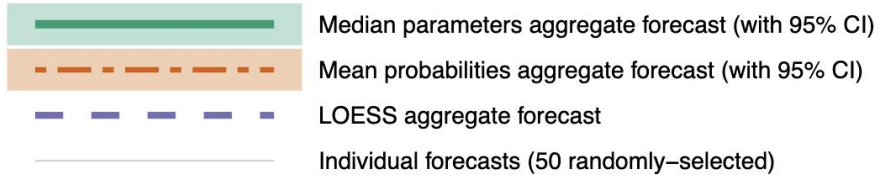


[21] [Zhang et al., 2022]

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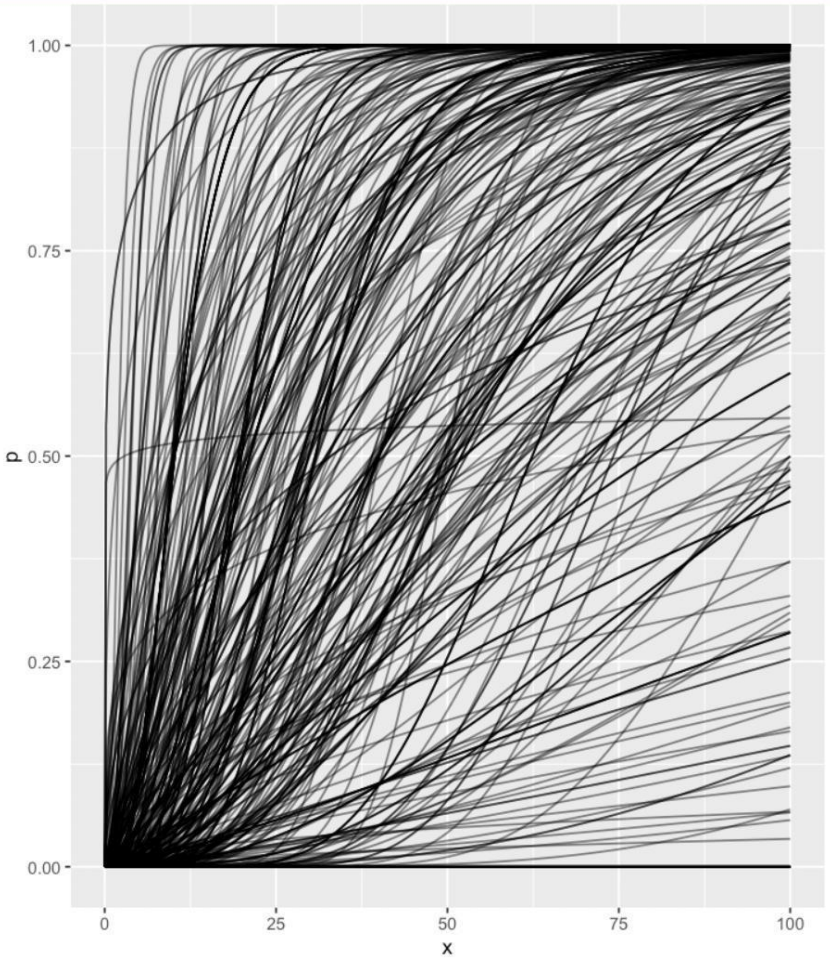


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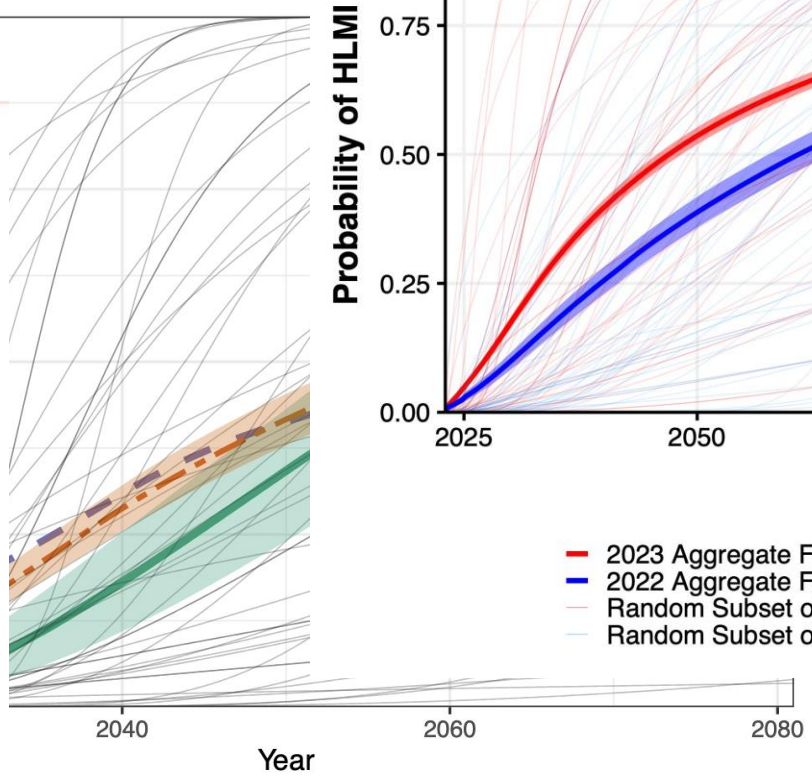


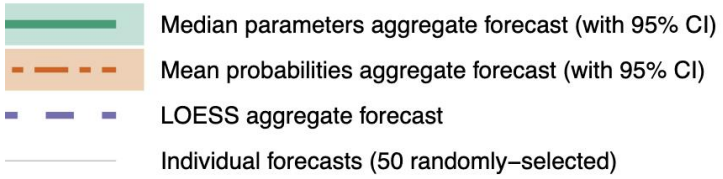
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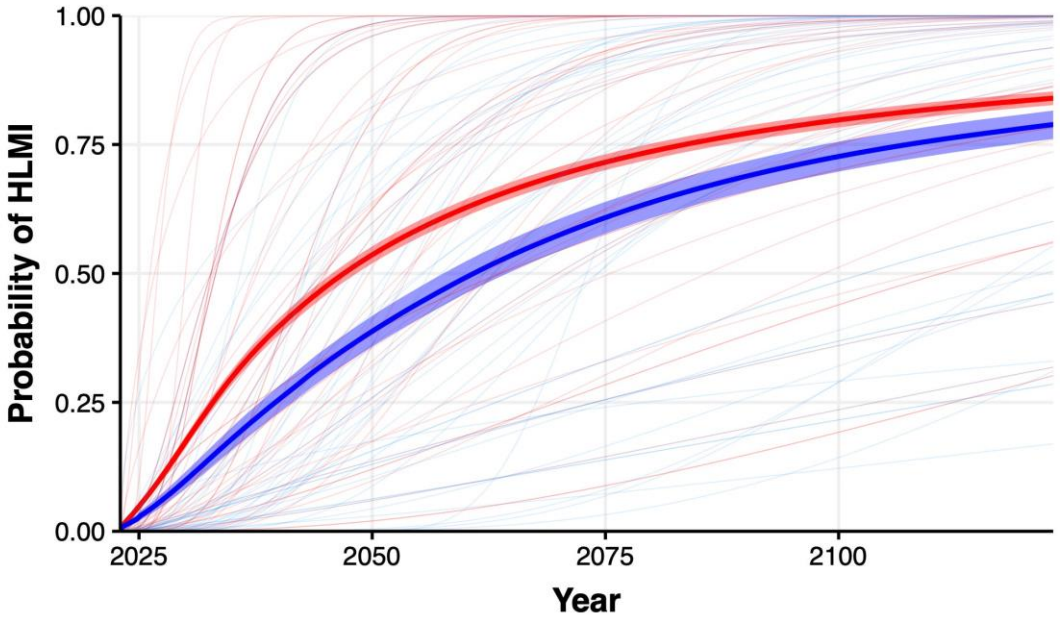


[23][Grace et al., 2022]



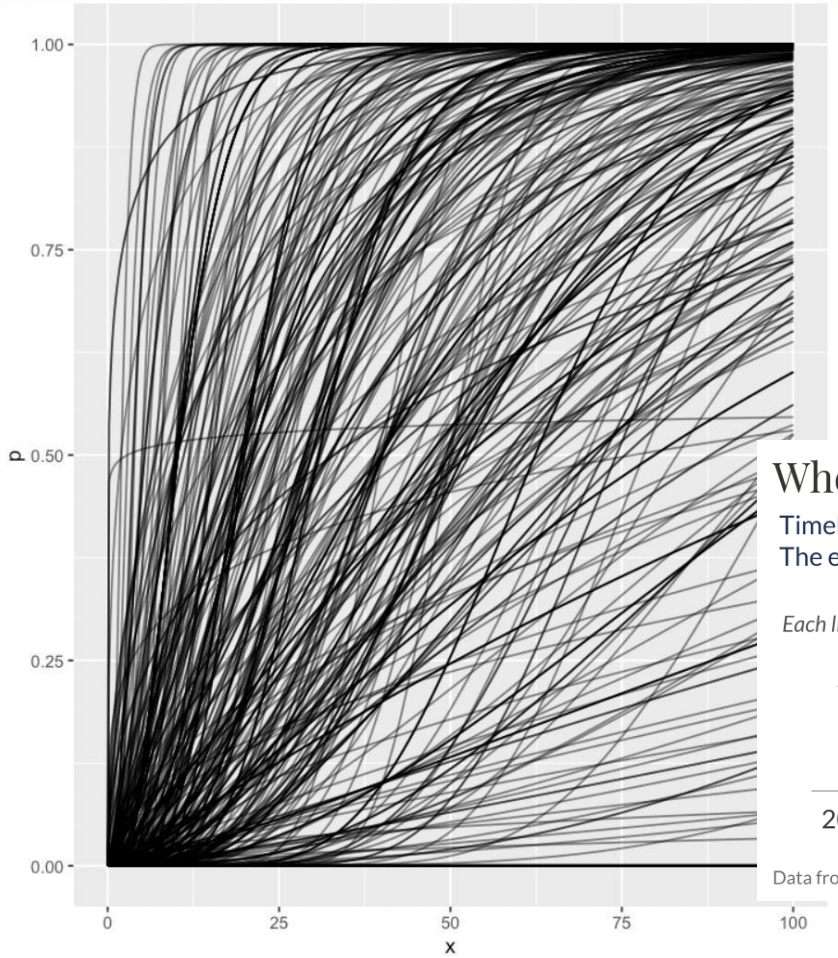


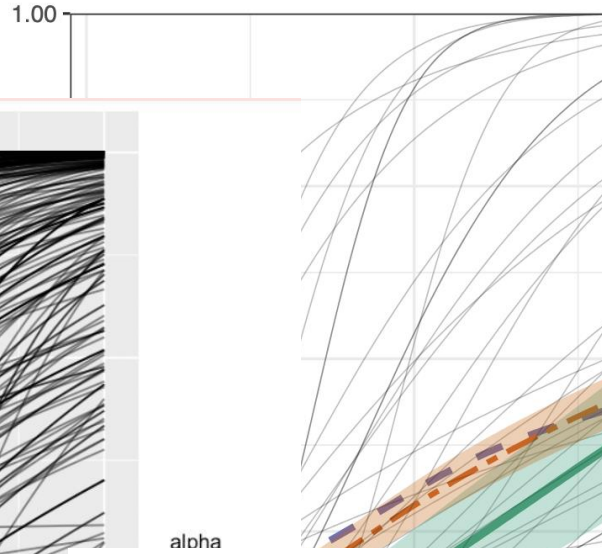
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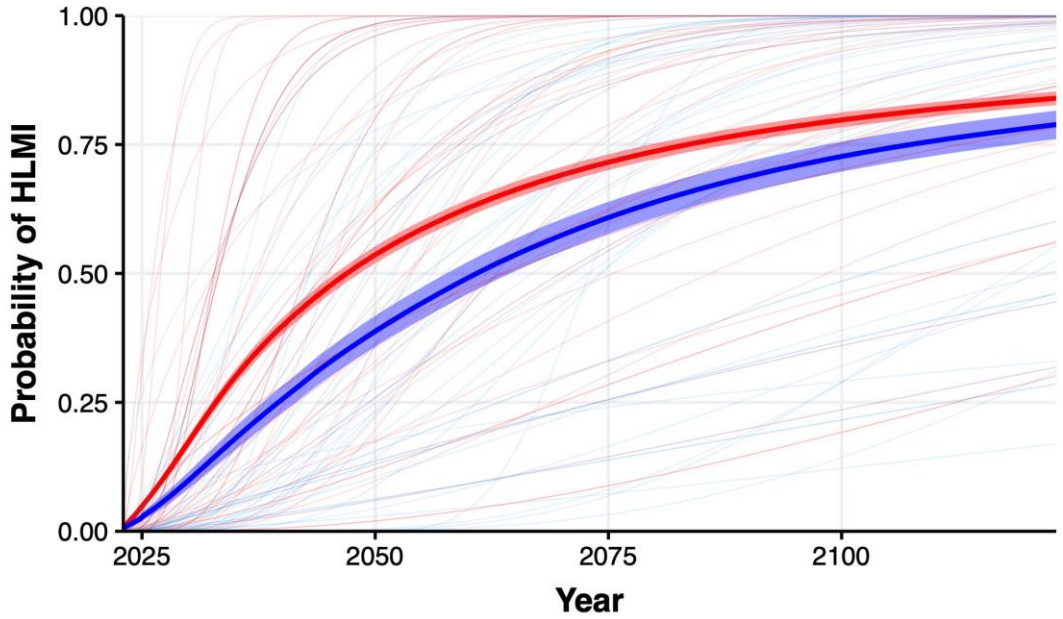


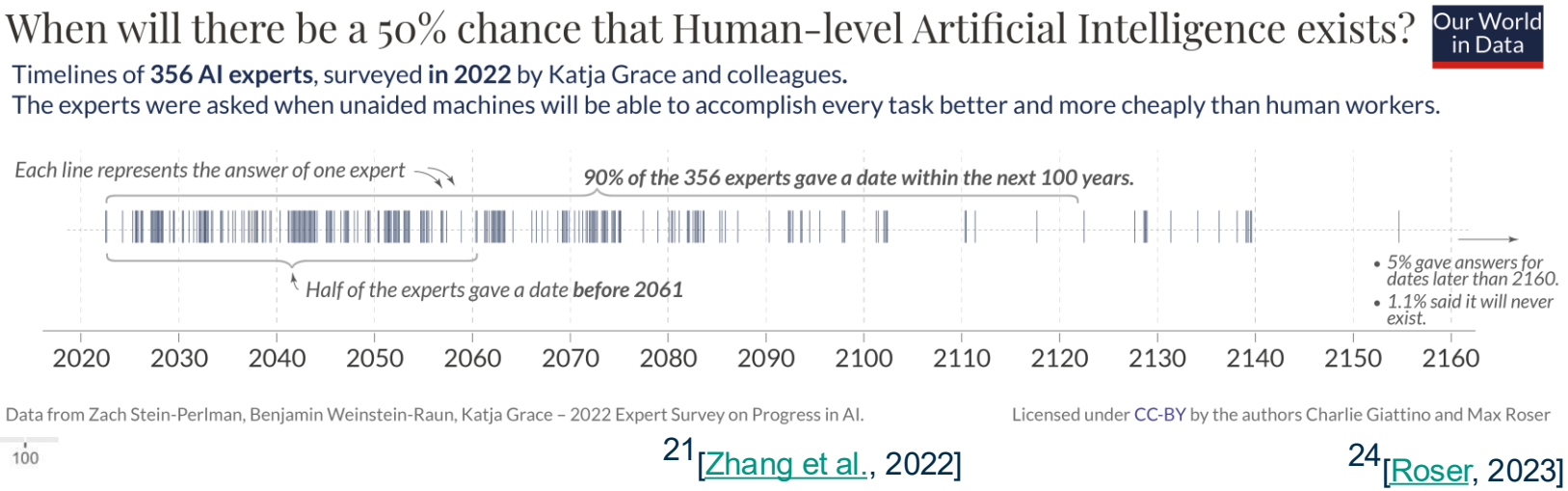
[22][Grace et al., 2023]

Feasibility - Timelines









[23] [Grace et al., 2022]

[21] [Zhang et al., 2022]

[24] [Roser, 2023]

Feasibility - Timelines

- AGI definitions differ
- “Experts” disagree
- Predictions change

Feasibility - Approaches

1. Advancing SOTA neural networks
2. Biologically inspired AI

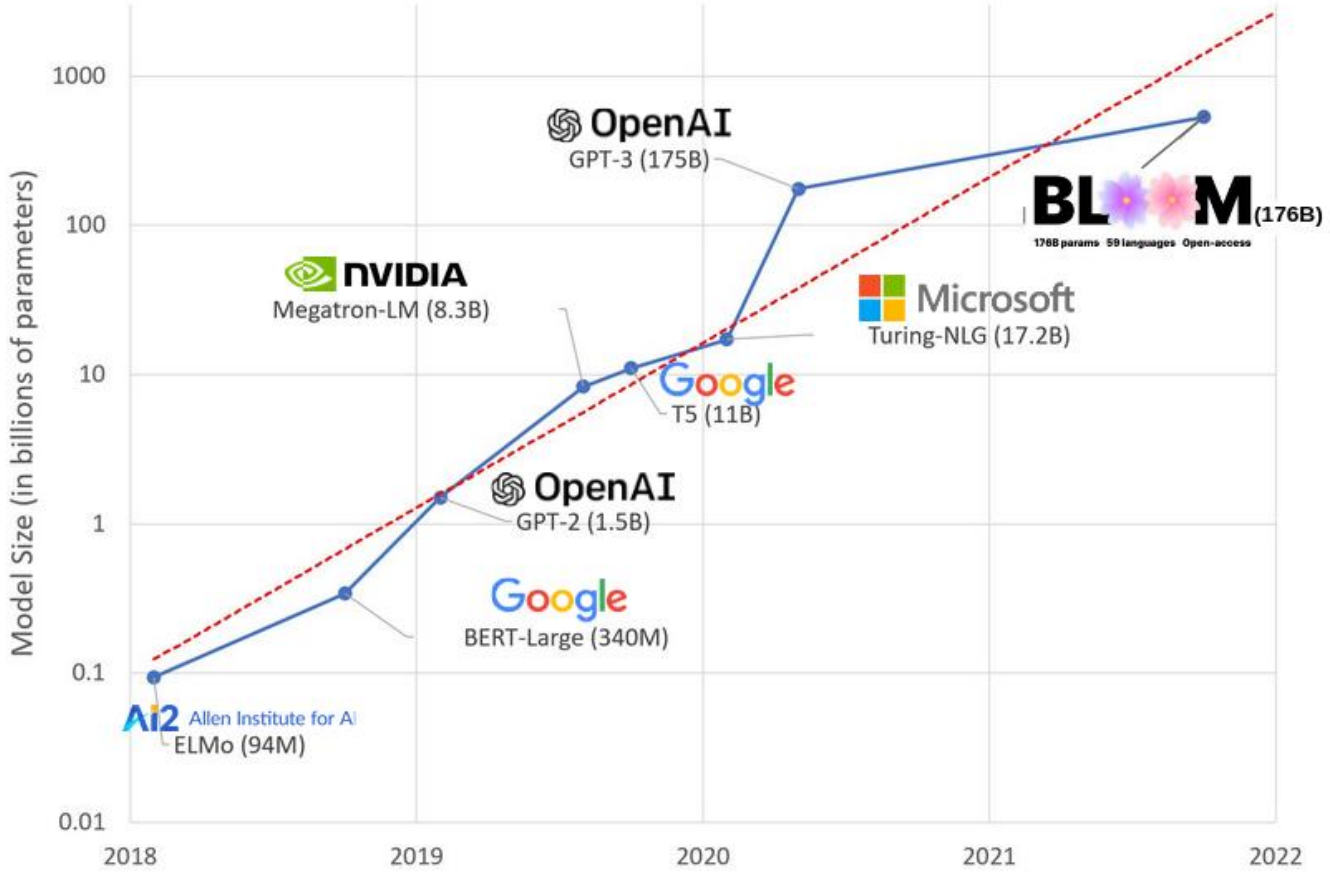
Feasibility - Better NNs

The approach:

- Larger networks
  - More complex architectures
- Increasing modalities[25]
- New learning techniques[26, 27]

The problem:

- Limited training data[28]
- Low data efficiency and generalizability[29]
- No biologic foundation[30]



[31] [ars technica, 2023]

[25] [Fei et al., 2022] [26] [Baydin et al., 2022] [27] [Ouyang, 2022] [28] [World Economic Forum, 2025] [29] [Huber et al., 2024] [30] [Sarfraz et al., 2025]

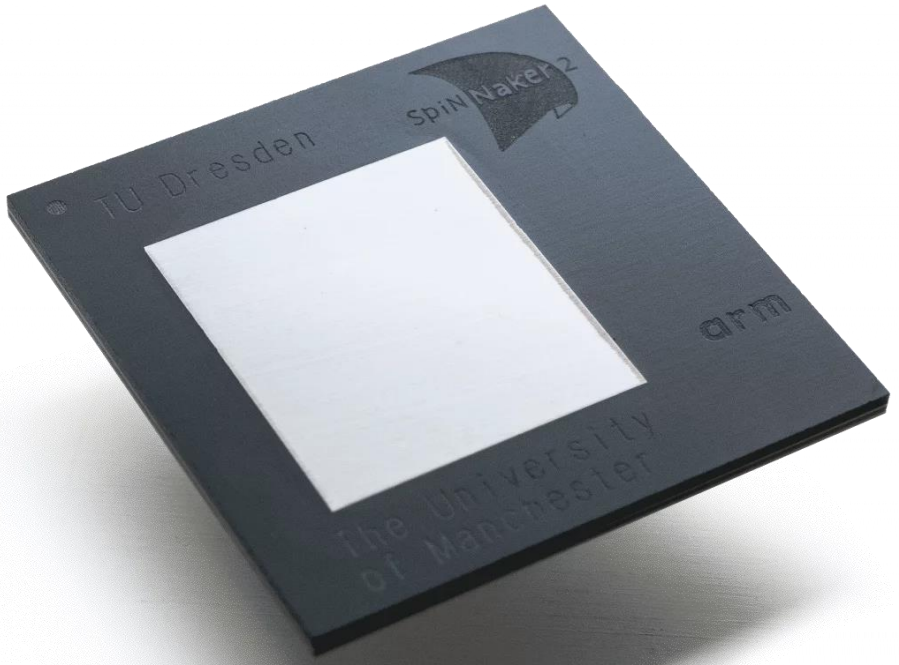
Feasibility - Bioinspiration

The approach:

- Neuromorphic computing
  - Spiking network architectures[32]
  - Specialized hardware[33]
- (Whole) brain emulation

The problem:

- Limited emulation power[34]
- Scanning resolution[34]
- Conceptual correctness[1]



[32][Maass, 2022] [33][Furber et al., 2022] [34][Zanichelli et al., 2022] [1][Russel & Norvig, 2020]

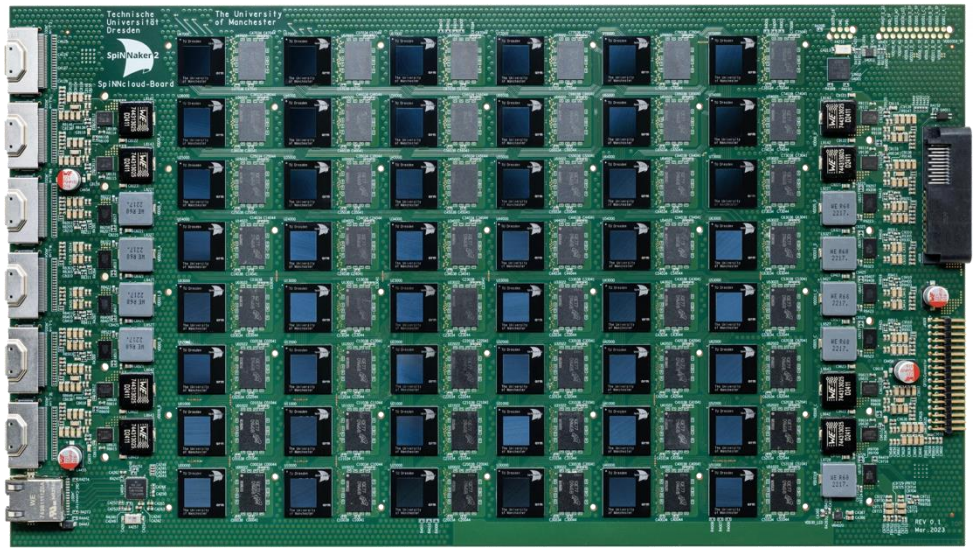
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[SpiNNcloud, Wikimedia]

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Implications

“AGI would have a big impact on several aspects of society”

Implications

Governance

Science

Employment

Ethics

Human Creativity

Privacy

“AGI would have a big impact on several aspects of society”

Economy

Security

Social Interaction

Education

Legal Systems

Implications - Prediction Accuracy

- Predictions are hard
- Increasing layers of assumptions
  - What is intelligence?
  - When is AGI achieved?
  - How does it function?
  - What properties does it have?

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How many and which assumptions are made?

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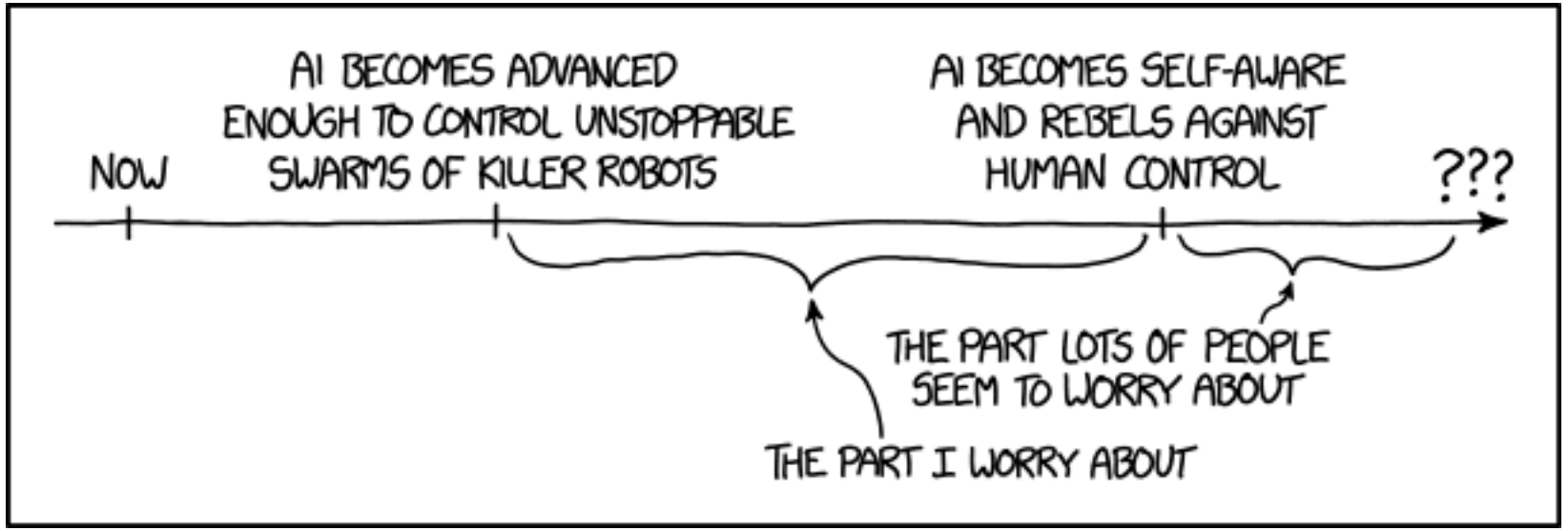
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- Intelligence is not well understood
- Ideas and concepts of AGI are diverse
- This kind of debate is neither new nor useless
- Important to focus on current AI developments



[xkcd 1968]

Discussion

- Which properties would you propose for an AGI benchmark?
- Do you consider any current technology a good candidate for a development into AGI?
- What are your thoughts on surveying "experts" on AGI development timelines and feasibility?
- Compared to other societal changes how do you evaluate human-equivalent AGI?

References

0[Life Magazine, 1970]

1[Russel & Norvig, 2020]

2[Hendrycks et al., 2025]

3[Hutson, 2025]

4[Turing, 1949]

5[Weizenbaum, 1966]

6[Fedorenko et al., 2025]

7[The Verge, 2025]

8[Phan et al., 1949]

9[Chollet et al., 2025]

10[arcprize.org]

11[Vice, 2015]

12[Goertzel et al., 2012]

13[Fan, 2025]

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16[Agüera y Arcas & Norvig, 2023]

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